Training of X300 Industrial Handling Robot

Auditor: Jian Lee Trainer: Alice Zhang Date: Nov 5th 2025

Training Content

1. How to deploy the robot?
2. How to set the screen APP?
3. How to use the iMS cloub platform?
4. Introduce the hardware.

Environmental Assessment

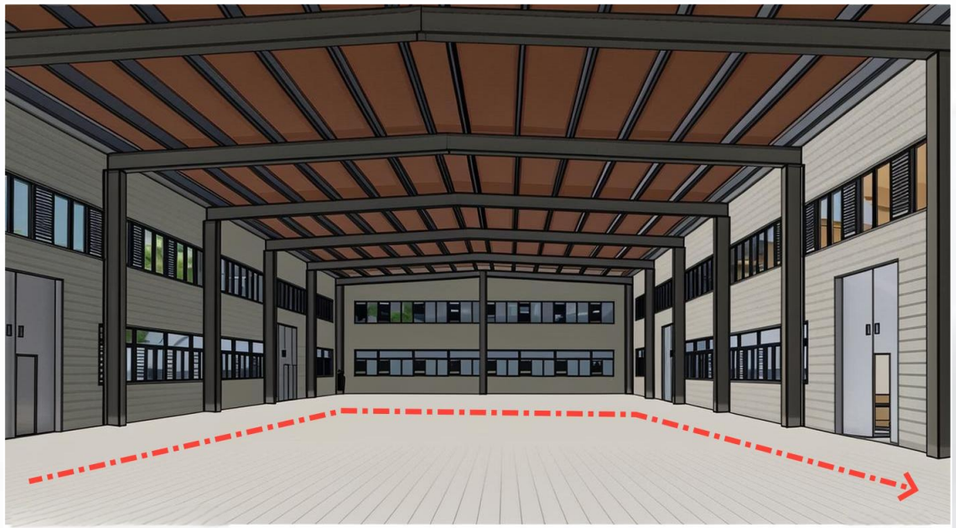
The robot uses single-line laser SLAM for autonomous positioning and navigation, enabling it to adapt to various complex indoor and semi-indoor environments. **To maximize deployment efficiency, it is essential to evaluate the environment in advance**, such as spatial layout, obstacle positions, lighting conditions, and ground environment. This will help optimize the robot's navigation path, reduce debugging time, and ensure smooth operation. The following scenarios may involve certain risks, so please pay more attention on them during deployment:

Open/Empty Scenes

- **Potential risks:**
  - Positioning errors may occur or positioning may fail entirely.

- **Causes:**
  - The robot’ s LiDAR scanning radius is **25 meters**; beyond this range, detection becomes ineffective, and error accumulation can easily lead to positioning issues.

- **Solution:**
  - Conduct a preliminary assessment to determine if the area is suitable for robot operation.
  - Operate the robot along the edges of the environment to ensure the LiDAR can stably scan wall structures.
  - Add other auxiliary positioning solutions (e.g., reflectors) after preliminary assessment.



For open/empty scenarios, scan the map along the edges (marked by red lines).

Environmental Assessment

High similar structural Scenes

- **Potential risks:**
  - Positioning errors may occur.

- **Causes of risks:**
  - Due to the high similarity of the scenes, errors may arise in the matching between laser point clouds and the on-site environment.

- **Solution:**
  - Conduct a preliminary assessment to determine if the area is suitable for robot operation.
  - Add other auxiliary positioning solutions (e.g., reflectors) after preliminary assessment.

A long corridor with no obvious structural changes on both sides

Environmental Assessment

Scenes with few features and high-dynamic

- **Potential risks:**
  - Positioning errors may occur or positioning may fail entirely.

- **Causes of risks:**
  - When the laser is installed at a horizontal height, the environmental features that the laser can scan are few, and the environment changes significantly. This leads to a large discrepancy between the current laser-scanned environment and the original mapped environment, resulting in positioning deviations.

- **Solution:**
  - Conduct a preliminary assessment to determine if the area is suitable for robot operation.
  - Add other auxiliary positioning solutions (e.g., reflectors) after preliminary assessment.

The goods are placed in large areas with high dynamic, and there are no other fixed and obvious positioning object for references.

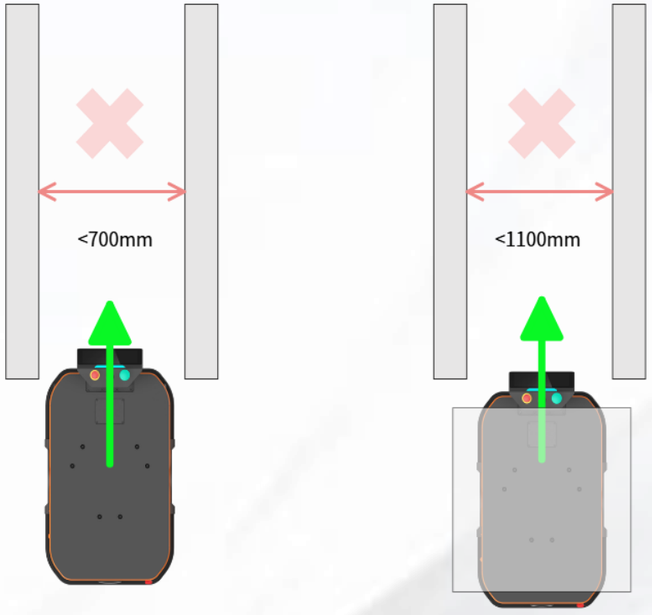
Environmental Assessment

Narrow and long passages

- **Potential risks:**
  - Possible scratches or collisions
  - Possible inability to pass through, or slow speed with low efficiency

- **Causes of risks:**
  - Obstacles that are too close may cause some obstacles to interfere with the robot's navigation.
  - Due to the detection effect of some obstacles and the influence of the passage shape, blind spots in perception may appear from the robot's perspective.

- **Solution:**
  Survey the passages in advance to avoid impassable ones and conduct a specific assessment of the handling plan.
  - X300 mode: 700mm forward, 1250mm rotation.
  - X300B-L shelf mode: 1100mm forward, 1350mm rotation (for 700 shelves).



The passage is too narrow for the robot to pass through

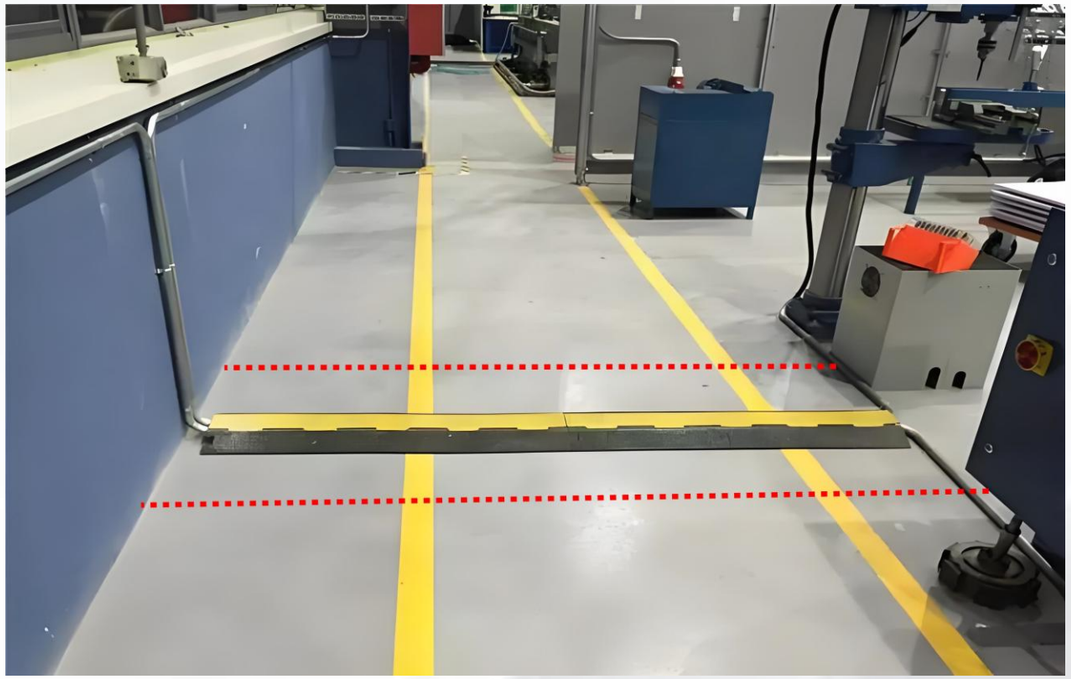
Environmental Assessment

Low-lying obstacles

- **Potential risks:**
  - May cause collision with low-lying obstacles

- **Causes of risks:**
  - The robot’s front detection system supports detecting obstacles >50mm in height.
  - Obstacles <50mm (such as wire ducts, forklift tines, and low pallets) may be undetectable.
  - The rear perception LiDAR for obstacle avoidance is installed at a horizontal height of 220mm.

- **Solution:**
  - Avoid routing the robot through areas with such obstacles.
  - Draw no-entry lines near obstacles to prevent the robot from planning paths into these areas.
  - Inform the customer that the robot’s operating routes must avoid such obstacles.
  - Additionally, the robot is equipped with a 360° full-enclosure collision strip.



Examples of obstacles <50mm, like the wire ducts and forklift tines.

Environmental Assessment

Gaps and steps

- **Potential risks:**
  - May be unable to pass through, get stuck, or trigger protection alarms

- **Causes of risks:**
  - The robot supports passing through gaps with a width of 30mm.
  - The robot supports passing through steps with a height of 15mm.

- **Solution:**
  - Small gaps and steps can be converted into gentle slopes.
  - Gaps and seams should be filled or covered with plates.
  - Avoid routing the robot through these areas.
  - Draw no-entry lines near such obstacles to prevent the robot from planning paths into these areas.

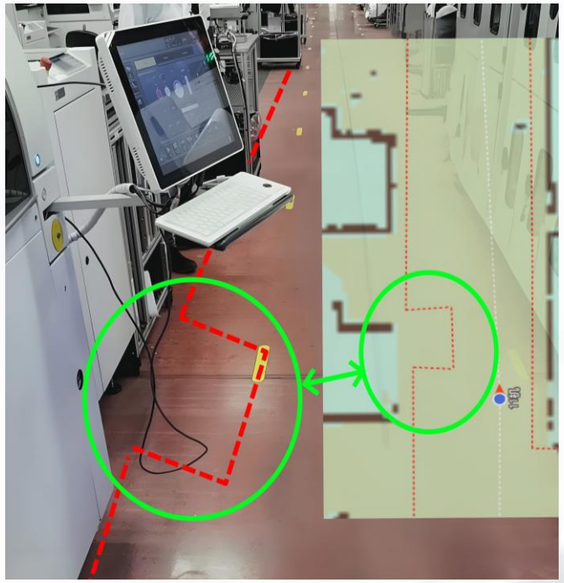
Environmental Assessment

Overhanging obstacles:

- **Potential risks:**
  - May fail to detect overhanging obstacles, leading to collisions or scratches

- **Causes of risks:**
  - The robot can detect front overhanging obstacles.
  - The rear perception relies solely on LiDAR for obstacle avoidance, which is installed at a horizontal height of approximately 220mm. When reversing or rotating, it cannot perceive overhanging obstacles at the rear or on the left/right sides.

- **Solution:**
  - Avoid routing the robot through areas with such obstacles.
  - Draw no-entry lines near obstacles to prevent the robot from planning paths into these areas.



1. For the suspended computer, please draw no-entry line at the outermost edge projection of the suspended computer.
2. For the suspended desktop, please draw no-entry line at the outermost edge projection of the suspended desktop.

Environmental Assessment

Falling hazards

- **Potential risks:**
  - There may be a risk of falling

- **Causes of risks:**
  - If positioning errors occur but the robot does not trigger a stop alarm, it may continue moving along the set route and potentially reach the edge of a platform.
  - When the speed is too high or the sensitivity of the inclined laser for detecting falling risks is insufficient, the robot may fall due to inertia.
  - The robot has no sensors to detect cliffs when planning backward movements.

- **Solutions:**
  - Avoid routing the robot through these areas.
  - Draw no-entry lines near such hazard points to prevent the robot from planning paths into these areas.

Unloading platforms, stairs, etc

Environmental Modification

Traditional environments were often not designed with robot applications in mind, so deploying robots in such environments may involve risks. To ensure the safe and stable operation of robots, it is necessary to assess and modify the environment.

Black/Highly reflective/Transparent/Thin poles

- **Causes:**
  - Lasers may fail to recognize black, highly reflective, transparent objects, or thin poles (< 20mm).

- **Solution:**
  - For obstacles undetectable by lasers (such as glass walls, ropes, and thin poles), affix stickers (made of materials with good diffuse reflection) at the laser height, covering a range of ±20mm.
  - Avoid operating the robot through these areas.
  - Draw no-entry line near such obstacles to prevent the robot from planning paths into these areas.

Transparent glass

Deployment Preparation

Tool backup list

● PC ● USB flash drives ● Multimeters ● Screwdrivers ● Hex socket sets, etc.

● Stickers for environmental transformation

Software: ● Burning tool ● Robot firmware ● Deployment tool

Unpacking Inspection

Due to the long transportation and storage time of the robot, the following inspections shall be conducted after unpacking:

Packaging Inspection: Confirm that the packaging is good without obvious extrusion damage. If there is any issues, take photos for evidence.

Accessory Inspection: After unpacking, check against the shipping list to ensure all accessories are completed.

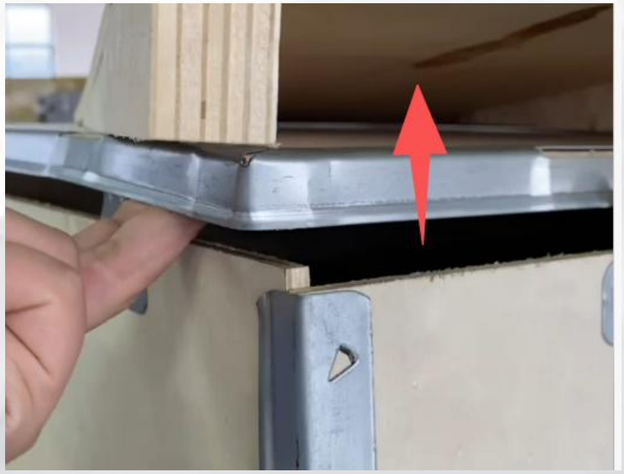
Function Inspection: Confirm that the robot can be powered on normally, complete the robot's self-inspection, and enter the main program.

Charging Inspection: Ensure that the charger works properly and the charging function is normal.

Deployment Preparation

Unpacking

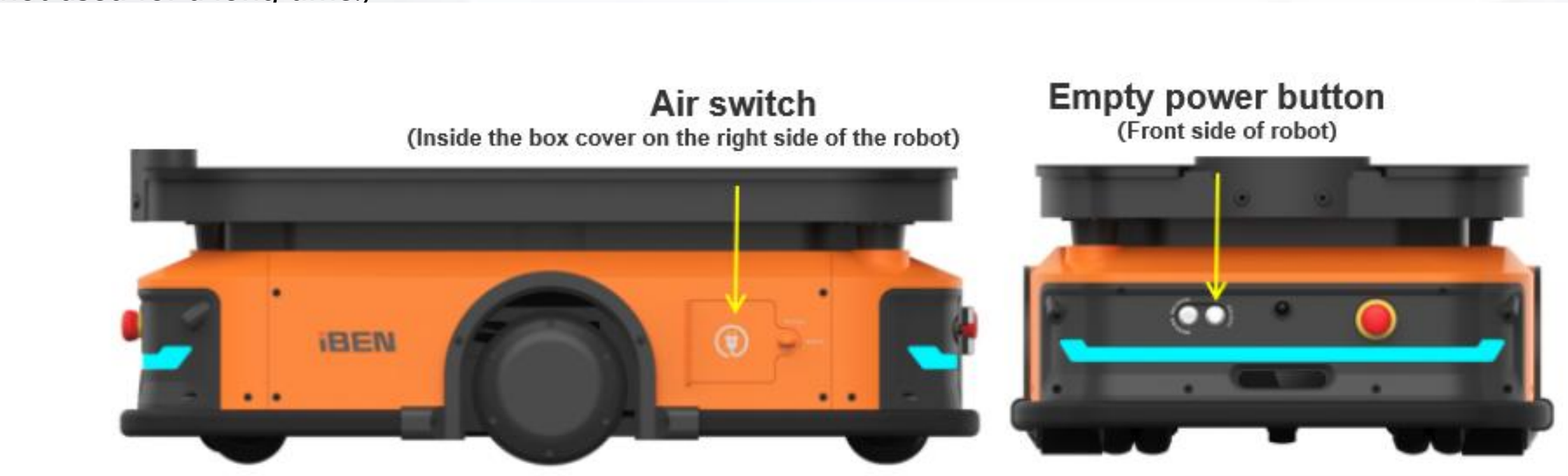
1. Please check if the packaging box is intact as shown in below picture. If confirmed intact, cut the strapping bands and remove all strapping bands and paper corner protectors.
2. After opening the box, as shown in below picture, please take off the upper EPE, then hold the load-bearing layer on the top of the chassis with both hands respectively to take the chassis out of the boxes.
3. Please take out the charging pile and power cable from the charging pile carton.



Deployment Preparation

Power-on/off

1. Turn the air switch to the “ON” position to turn on the power supply of the robot.
2. Power-on: Press the power button briefly. When the indicator light of the button turns on, the chassis light strip and the interactive screen turn on, it means the power-on is successful.
3. Power-off: Press and hold the power button until the indicator light of the power button goes out, the light strip and the screen turn off, it means the power-off is successful.
4. Turn the air switch to the “OFF” position to turn off the power supply of the robot. (Please turn off the power if the robot is not used for a long time.)



Deployment Preparation

Supplementary Charging

After the robot is stored for a long time, the battery may be low power. In this case, the robot should be charged timely to ensure sufficient battery power to support subsequent deployment.

Version Confirmation

It is necessary to check whether the software tools have been updated to the latest version. It is recommended to perform this check before the machine arrives at the site to avoid troubles caused by historical issues.

Create map and points

To achieve the function of autonomous positioning and walking for material handling, the map-scanning and point-building robot needs to create a map. Therefore, when use the robot for the first time, create the map first, edit the map, and then create points on the map.

Charging Pile (Point)/Initial Point

To simplify the operation process, the "initial point" and "charging pile" can be unified into the same coordinate. Start mapping from the charging pile, which serves as the reference origin of the global map; During each subsequent power-on, the charging pile point is also used as the reference for positioning initialization. To ensure reliability and consistency, the position of the charging pile must be determined and standardized in advance before powering on and scanning the map, so that it can meet the dual requirements of charging and positioning initialization simultaneously.

Charging Pile: It is the automatic recharging point of the robot and the repositioning point of the robot's charging pile.

Initial Point: It is the starting point for map creation. If the robot is powered on not at the bound charging pile, the robot will use this point as the repositioning point.

Requirements for Placing the Charging Pile

1. Left and Right Clearance: A horizontal clearance of more than 450mm shall be reserved on both sides to avoid restrictions on charging docking and leaving the pile.
2. Securing the Charging Pile: It can be fixed on the ground with ground nails or placed it against a solid wall for stability.
3. Ground Conditions: Don’ t place the charging pile on carpets, slopes, steps, or locations with obvious height differences, so as to prevent "slipping" or docking failure caused by height differences.
4. Position Marking: The position of the charging pile shall be clearly marked and numbered to prevent failure to accurately restore its position after being moved.
5. Front Clearance: The straight passage in front of the charging pile shall be more than 1800mm, so as to ensure the feasibility of the robot's entry and exit docking path and provide a safety buffer.

Charging Pile (Point)/Initial Point (Opposite Side of the Point)

1. Structural Features: Please select area with obvious geometric features, such as wall corners, columns, and walls with clear structure, so as to improve the stability of algorithm matching.
2. Environmental Stability: During mapping and subsequent operation, environment should not change frequently; Please avoid the large-scale temporary stacking, frequent renovations, or furniture adjustments.
3. Space Selection: Avoid transparent glass, long straight corridors, Open large spaces, and completely cluttered areas to reduce positioning drift caused by laser penetration, repeated and sparse features.

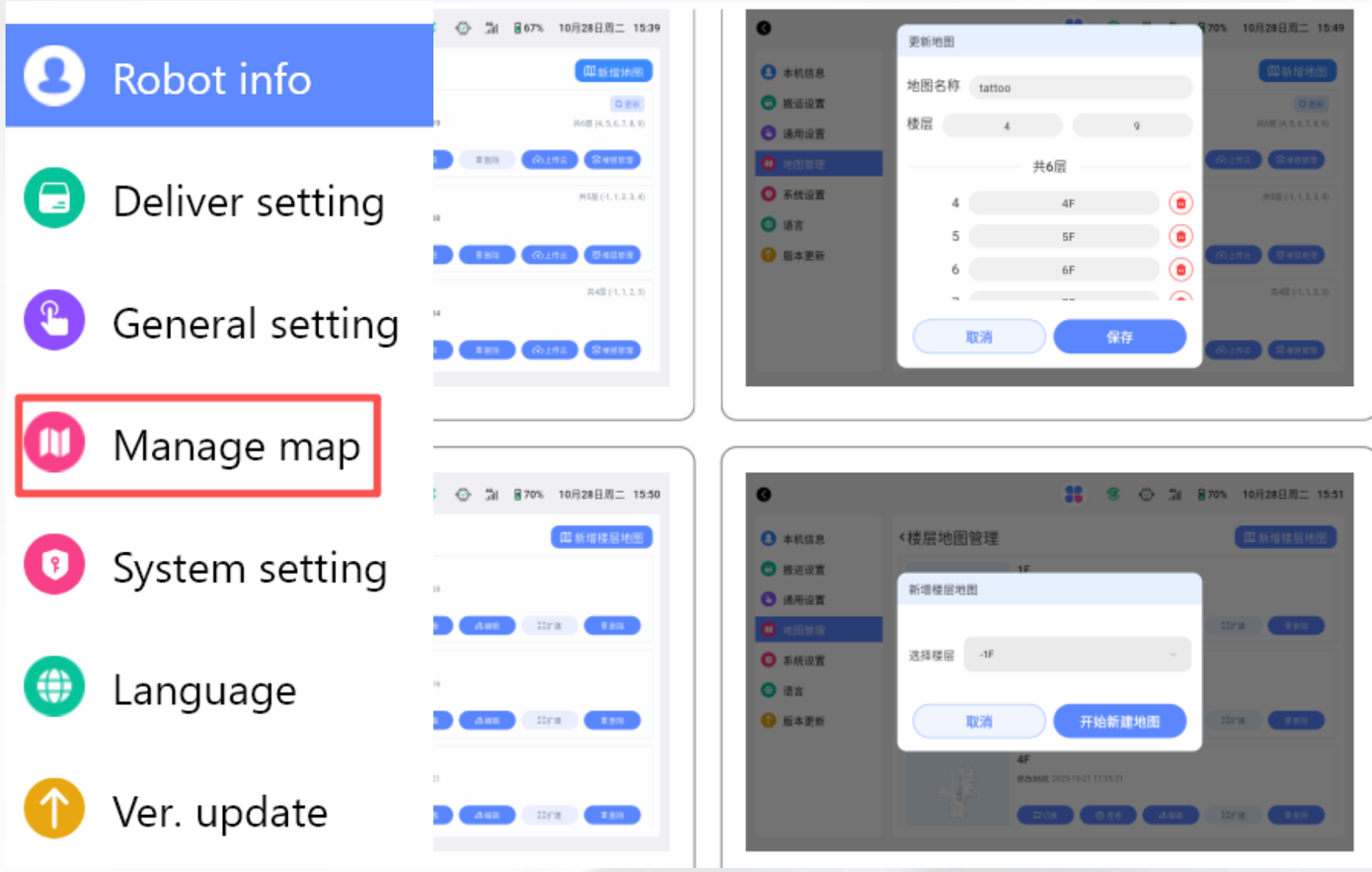
Map Scanning

The robot's map scanning can be implemented using a variety of deployment tools, including Vivid, the robot cloud-based iMS platform, and the interactive screen APP. You can refer to the specific user manual of each tool for map scanning and deployment. The following is a brief guide to mapping via the interactive screen APP.

Mapping Guide of robot APP

1. Keep the robot charging at the charging pile.

2. On the home page, click "**Settings**", enter the password “**6666**” , then select "Manage map" and "Add a New Map". Input the map name and the floor of the building corresponding to the map. Locate the "Floor Management" option for the newly created map, add a floor map, select the corresponding floor, and click "Start Creating New Map".

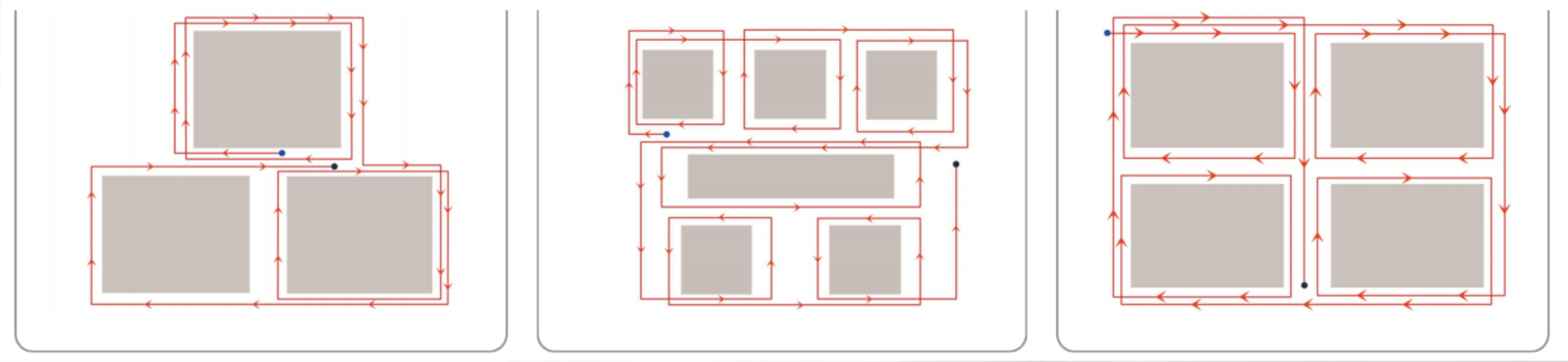


Start Creating a New Map

According to the illustration, "Start Creating a New Map", press the parking button, push the robot to scan the areas that robot will work one by one, and save the map after completion.

Key Points for Mapping

1. Try to scan the scenarios when there are no people or only a few people in the area.
2. Please remove objects that frequently move in the area.
3. When start mapping, push robot to move at an average speed, properly, smoothly and straightly to avoid winding routes.
4. Please push the robot to scan all areas where the robot operates.
5. If all obstacles have been fully scanned, there is no need to scan repeatedly.
6. Prioritize the principle of small closed loops and large closed loops when creating the map.
7. **After scanning all areas, push the robot return to the initial point (charging pile) of map creation.**



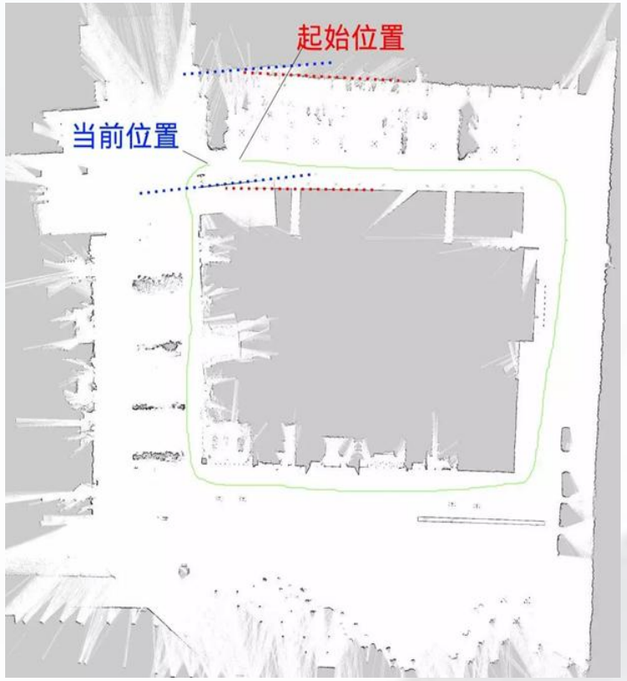
Common Issues and Solutions for Mapping

After mapping, the accuracy of the map directly affects the robot's positioning and navigation performance. To ensure the robot operates safely and efficiently in real environments, please check the map strictly and confirm it meets quality requirements.

Normal loop closure: For circular environments, please ensure that the circular closure of the map is consistent with the real environment.

Inspection method: Check the circular part of the map to see if there are any misalignments or crossings.

Solutions:
1. If the loop closure of a finished map is abnormal, please re-scan the scenarios.
2. For maps involving circular paths during scanning, after completing a full loop, continue moving a few more meters to observe the loop closure. The backend loop closure algorithm will perform matching and properly handle the loop closure.



Common Issues and Solutions for Map Scanning

Ghosting: There should be no multiple same objects or structure in the map; all scanned objects should appear only once on the map.

Inspection method: Carefully check the map for duplicated structures or objects, such as walls, columns, or obstacles appearing twice or more. If ghosting occurs, it may indicate errors in the robot's positioning or data processing, and a reassessment is needed to determine whether re-scanning is required.

Solutions:
1. For minimal and slight ghosting, the map can be repaired through editing.
2. If ghosting appears over a large area, please re-scan the scenarios. During re-scanning, please check whether the robot's laser and other sensors work normally (e.g., tilted laser installation, abnormal chassis odometry, etc.).

Common Issues and Solutions for Map Scanning

Incomplete map: For areas where the robot operates, all key regions should be scanned.

Inspection method: Compare the map with the actual site to ensure all areas where the robot will operate are included in the map.

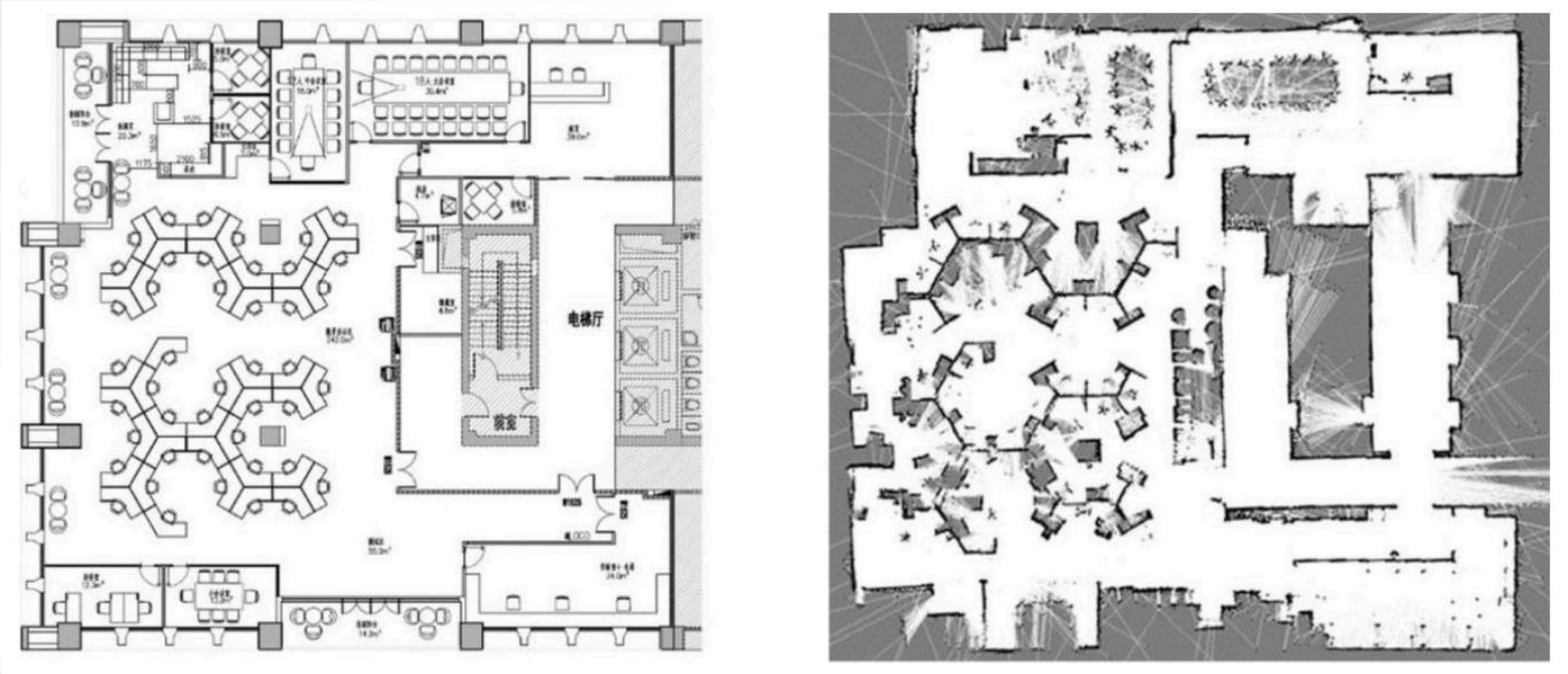
Solutions:
1. During scanning, push the robot to scan all areas where it works.
2. If an incomplete map is found after scanning and no other abnormalities are detected, the map can be supplemented by extending it.

Common Issues and Solutions for Map Scanning

Consistency with the real environment: The map should accurately reflect the layout of the real environment, with all walls, obstacles, passages, etc., matching the real site.

Inspection method: Compare the on-site layout with the map item by item to ensure every detail on the map, including spatial proportions and object positions, is consistent with the actual environment.

Solutions:
1. Check the robot's status and re-scan the map.



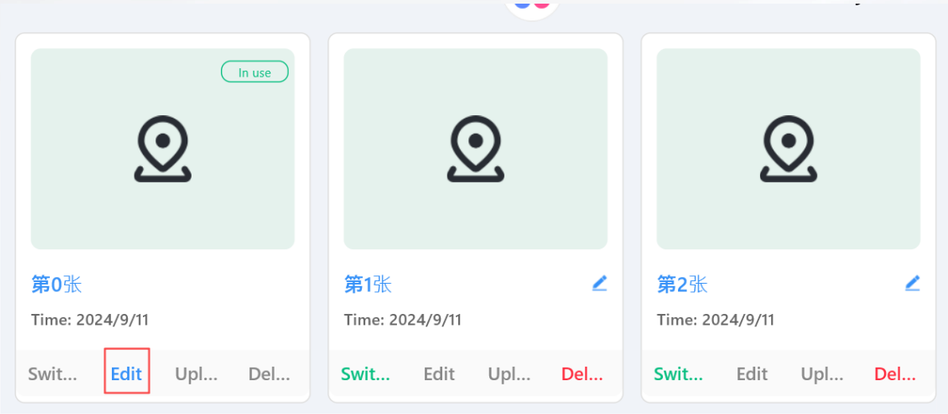
Create Point

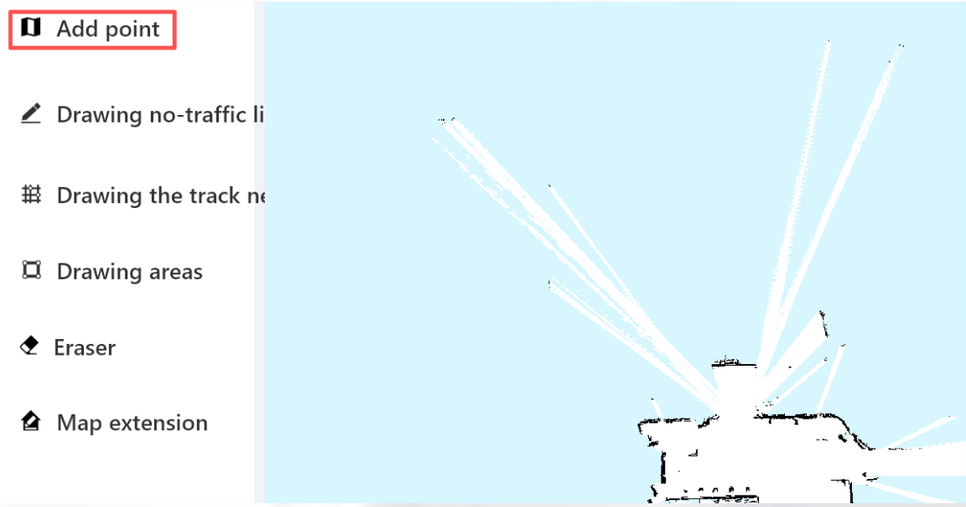
After creating the map, please add the points. This can be achieved through various deployment tools, including Vivid, the iMS robot cloud platform, and the interactive screen APP. You can refer to the corresponding manual as needed.

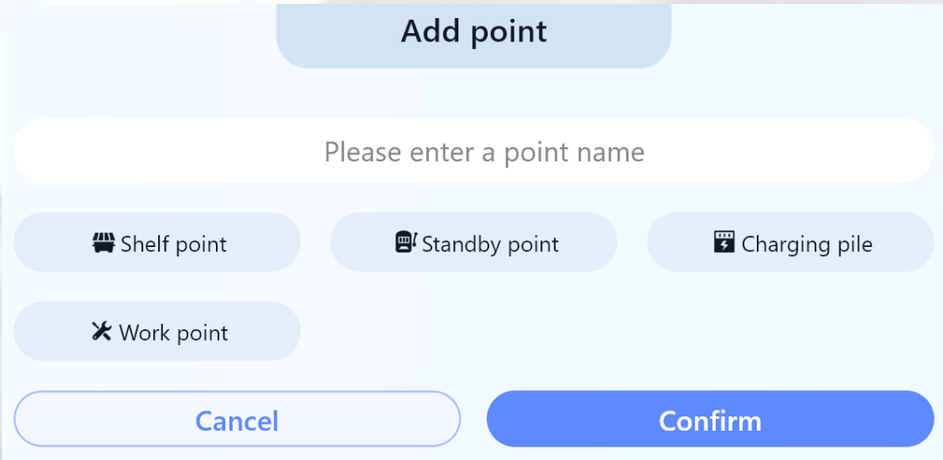
Add Points APP Quick
Enter the "**Map Editing**" interface, select "**Points**", then choose "**Add Current Point**" or "Add Designated Point".

Add Current Point (Recommended):
- Push the robot to the position where a point needs to be added.
- Click "Add Current Point".
- Enter the point name (it is recommended to use a short name that is easy to distinguish and remember).
- Select the point type.
- Confirm to complete the addition of a point.

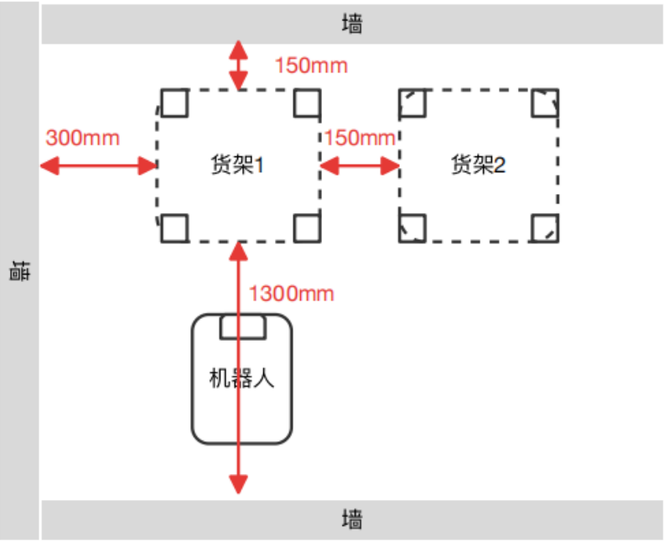
Add Designated Point:
- Click "Add Designated Point".
- Enter the point name (it is recommended to use a short name that is easy to distinguish and remember).
- Select the point type and confirm.
- Drag the map, click on the map, and a point will be generated at the clicked position.







Explanation of Point Types



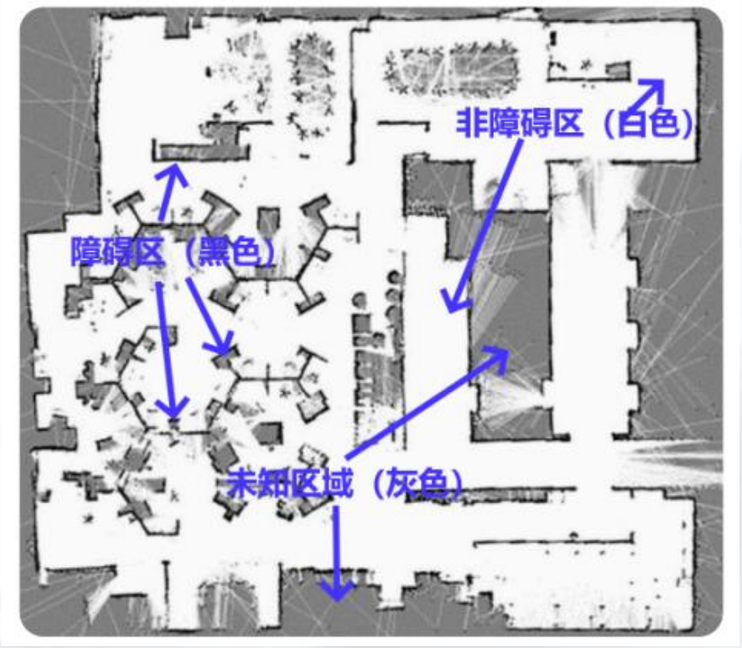
Shelf Point: A point where the robot, after reaching the designated position, needs to identify the shelf, then dive into the shelf for lifting or place the shelf.
Standby Point: A point where the robot stays idle.
Charging Pile Point: The automatic recharging point of the robot and the repositioning point of the robot's charging pile.
Initial Point: The starting point for map creation. If the robot is powered on not at the bound charging pile, the robot uses this point as the repositioning point.
Work Point: A point to which the robot navigates.

Shelf Point:
1. Place the shelf at the required position, pay attention to the orientation of the shelf, and reserve space for the shelf according to its specific size (the one on the left is a standard **700*700mm shelf**).
2. Manually push the robot into the bottom of the shelf, center it within the shelf, add a shelf point, and complete the addition of a shelf point.
3.After the shelf is placed, a mark can be pasted on the ground as the fixed position of the shelf point.

Edit Map

After creating the points, please edit the map, including editing the original positioning map, drawing no-entry lines and zone, and adding map areas. This can be achieved using a variety of deployment tools, such as Vivid, the iMS robot cloud platform, and the interactive screen APP. You may refer to the corresponding manual as required.

Explanation of Map Elements



Non-Obstacle Area (White)
The white part on the map, it indicates areas in the environment where robot can move. Generally, most of a map consists of non-obstacle areas.

Obstacle Area (Black)
The black part on the map, it indicates obstacles in the environment, such as walls, vehicles, or office desks and chairs. Meanwhile, lasers are used for positioning in this area.

Unknown Area (Gray)
The gray part on the map, it indicates areas in the environment where sensor can not detect, and the safety status remains unknown.

Edit Map

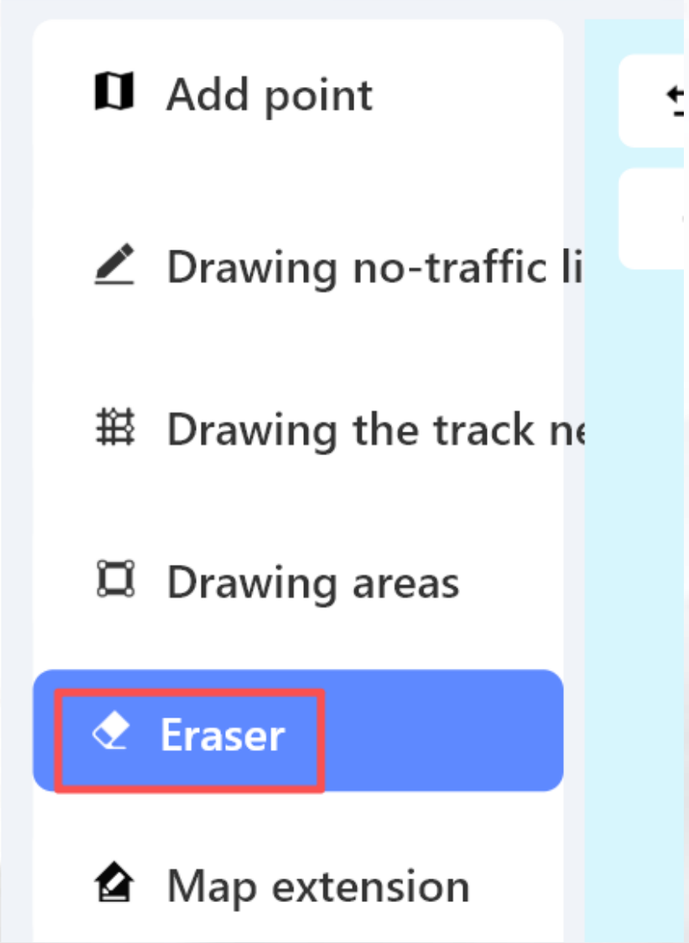
Creating the map based on the real environment. Use the eraser tool to remove temporary objects generated during creating map (such as people and temporary obstacles), so as to restore the information of real obstacles on the site. This helps improve the robot's positioning and path planning efficiency.

Eraser Tools:

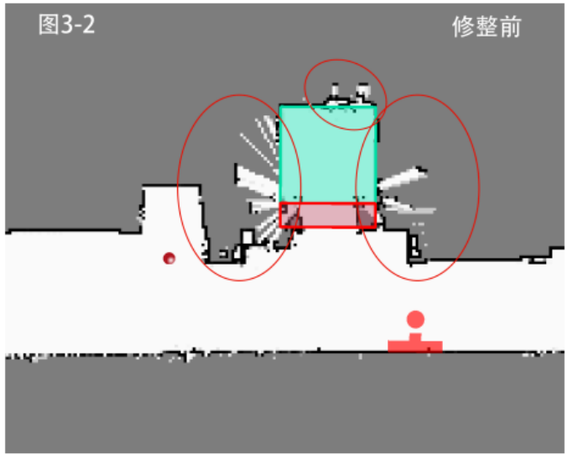
1. White Eraser: Generally, fixed equipment such as walls and machines are shown as black features on the map. The white eraser can remove other movable equipments and temporary objects (caused by temporarily placed obstacles scanned into the map by the laser).

2. Gray Eraser: It is used to erase burrs in the scanned map caused by point cloud penetration when scanning on-site objects like glass.

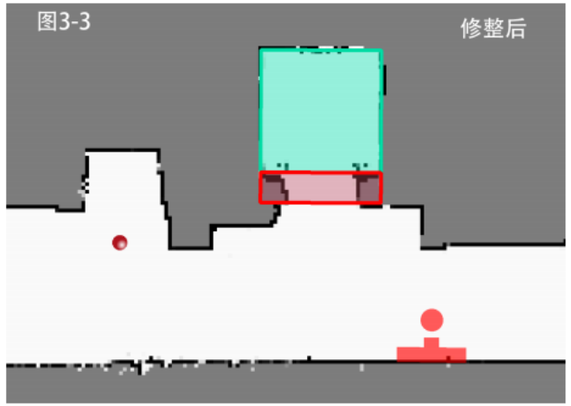
3. Black Eraser: For objects that actually exist during mapping but are not reflected on the map, or when the entity has poor reflectivity, the black eraser can be used to manually add objects to enhance positioning stability (use with caution).



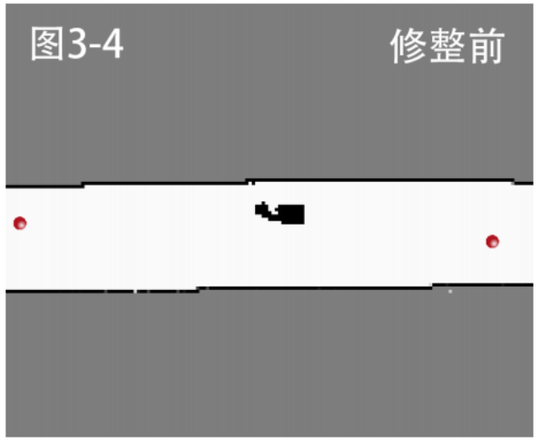
Eraser Tools



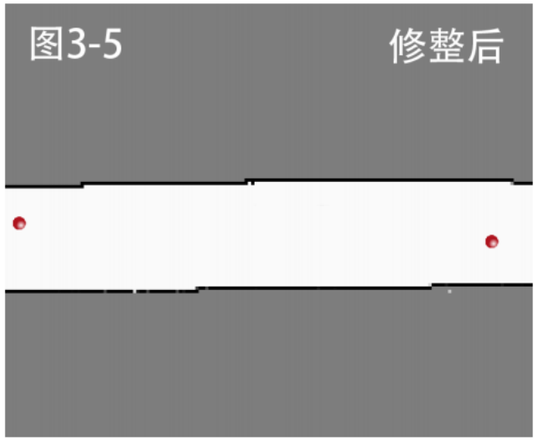
噪点擦除前



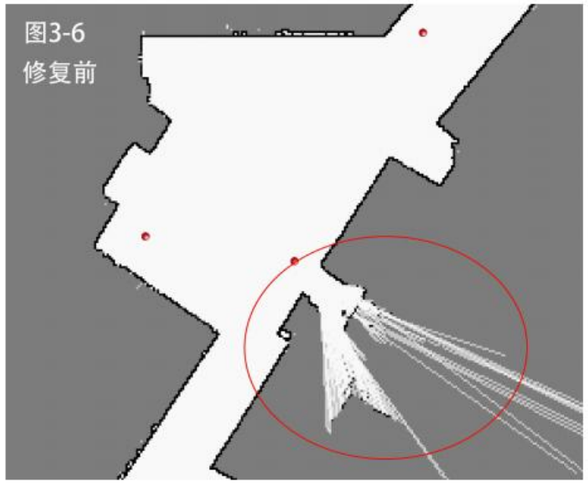
噪点擦除后，非必需



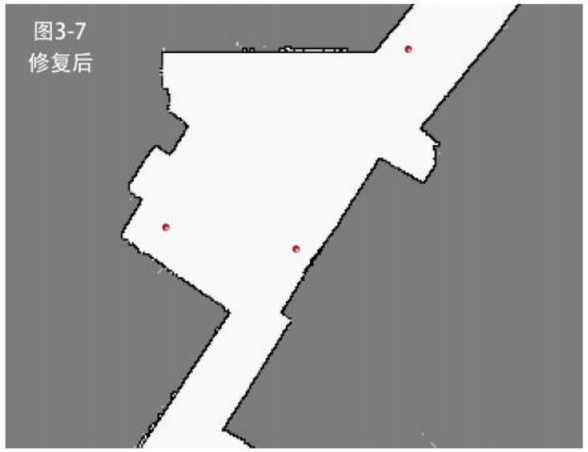
动态特征擦除前



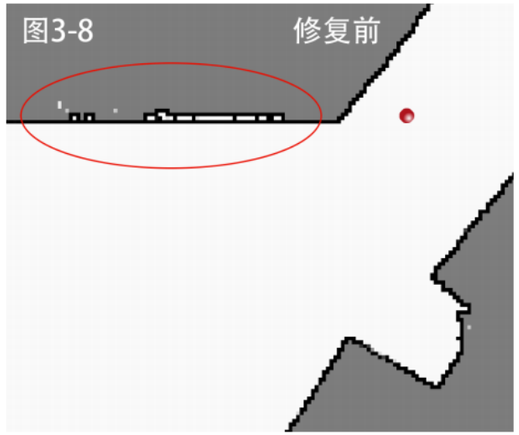
动态特征擦除后



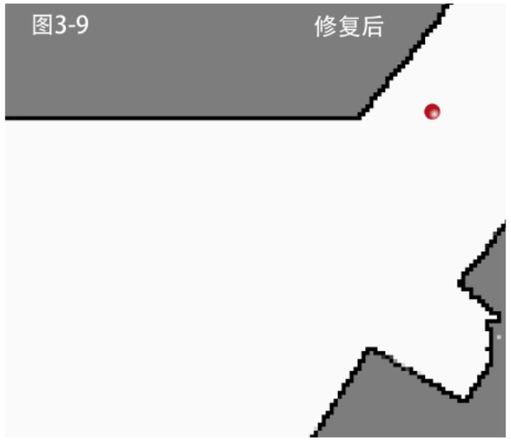
扫图时开门



常规使用关门，非必需，可按需编辑修复



重影消除前

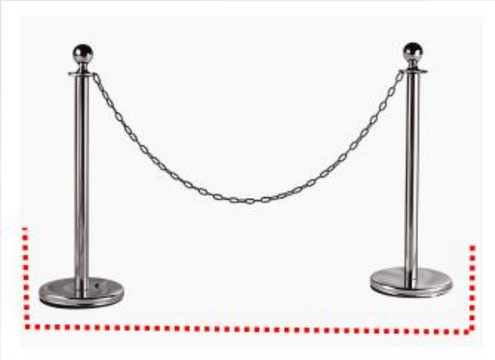


重影消除前，非必需

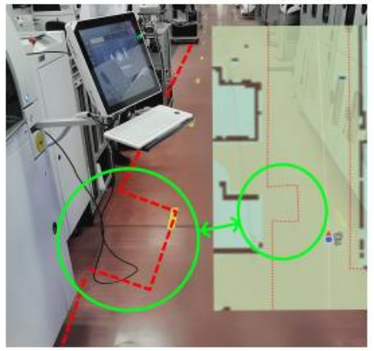
Draw No-entry Line and zone

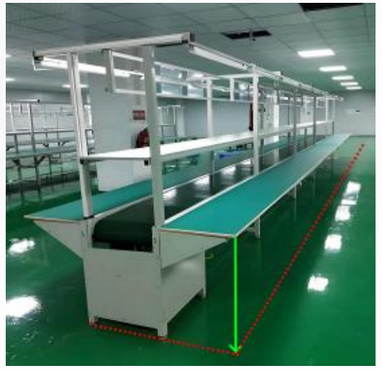
During the deployment of industrial robots, the drawing of no-entry lines is crucial for ensuring robot safety, avoiding collisions, and optimizing path planning. By setting these areas on the map, the movement range of the robot can be effectively restricted, preventing it from entering dangerous areas or making misoperations.

Zone Editing: Deceleration zone: Speed-limiting area, where the robot's speed is restricted after entering.
No-entry zone: Prohibit robots from entering, with the same effect as the no-entry line



Suspended fence



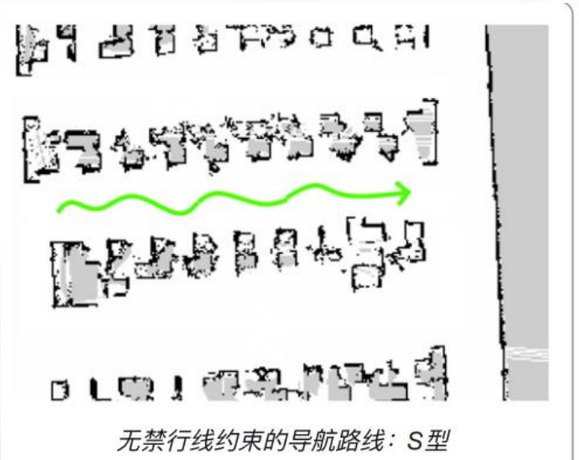


Suspended computer and line

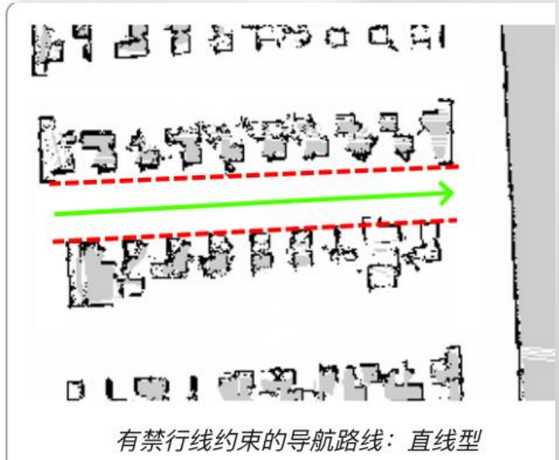
Unloading platforms, stairs, etc

Low-lying wiring duct

Transparent glass



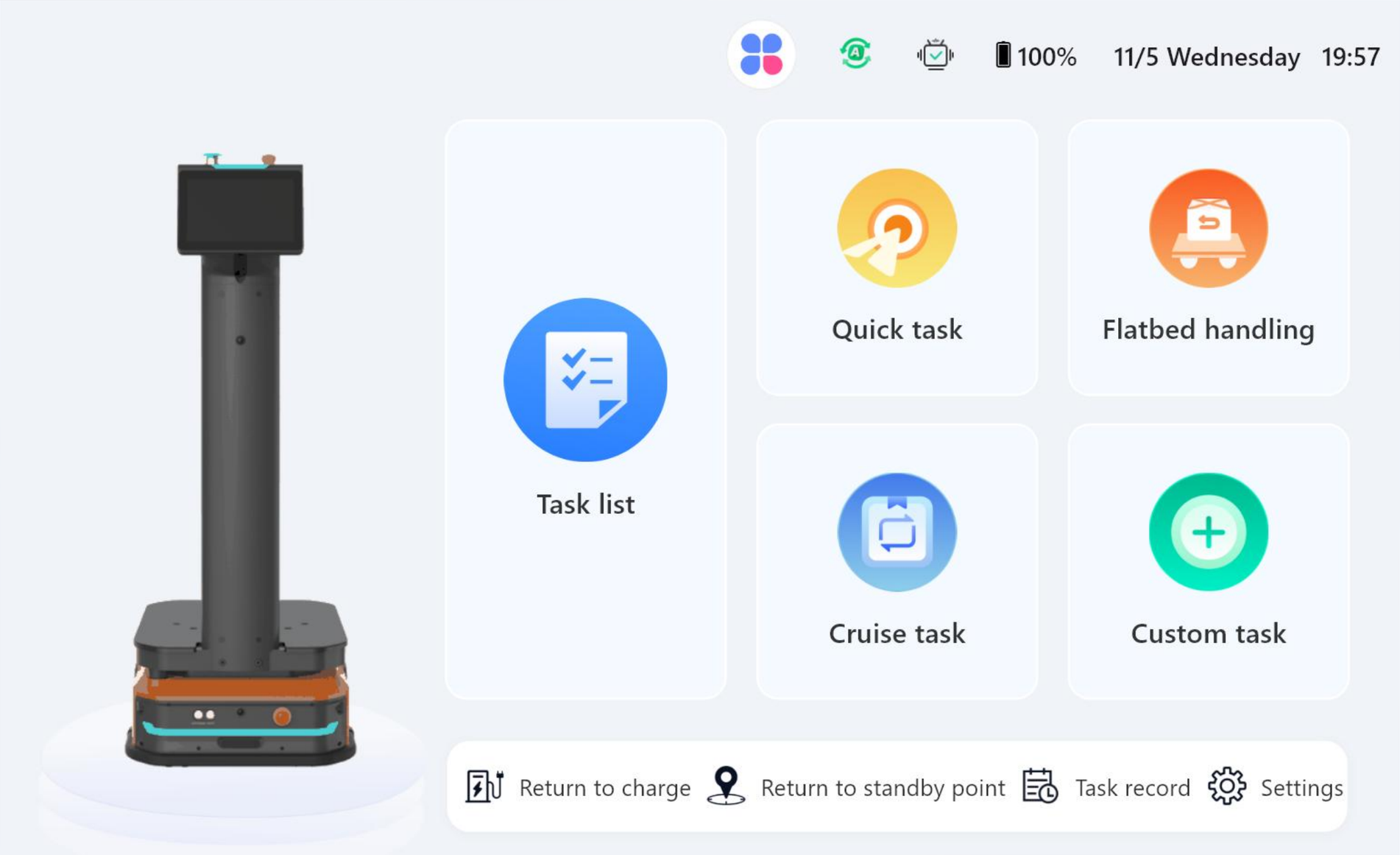
无禁行线约束的导航路线：S型



有禁行线约束的导航路线：直线型

Task configuration

1. In order to adapt to different industrial scenarios, the X300 robots provide users with a variety of task modes. After the robot is turned on, users can choose task modes according to different needs on the home page.



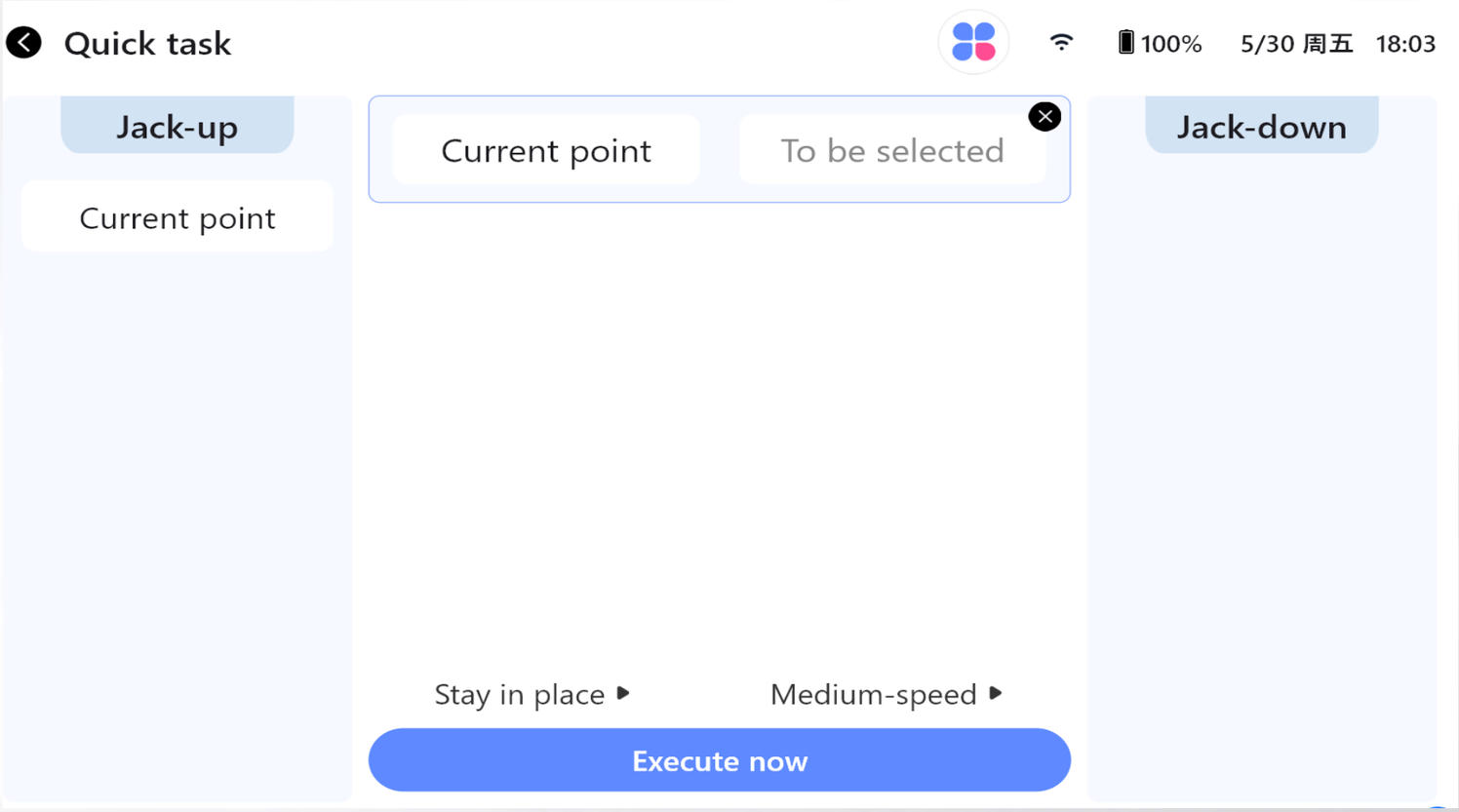
Task Description

Task Modes	Description
Quick task	Users can set jack-up and jack-down points. The robot will automatically move to the jack-up point, identify the shelf, lift the goods, transport them to the destination, and automatically unload the shelf.
Cruise task	Users can set a route, and the robot will run along the route for a set number of times. It can stop at designated points on the route for a specified duration, facilitating the pickup and delivery of goods during patrol. After task setup, it will be automatically saved in the task list, making it easier to execute the task quickly.
Flatbed handling	Enables convenient configuration of single-point-to-multi-point flatbed handling tasks. This task mode does not support lifting.
Custom task	Users can set a route, and the robot will run along the route for the set number of times. It can jack-up and jack-down shelves at designated points on the route, stop at points for a specified duration. After task setup, it will be automatically saved in the task list, making it easier to execute the task quickly.
Task list	Allows viewing, editing, starting, and deleting the cruise tasks and custom tasks.

Quick task

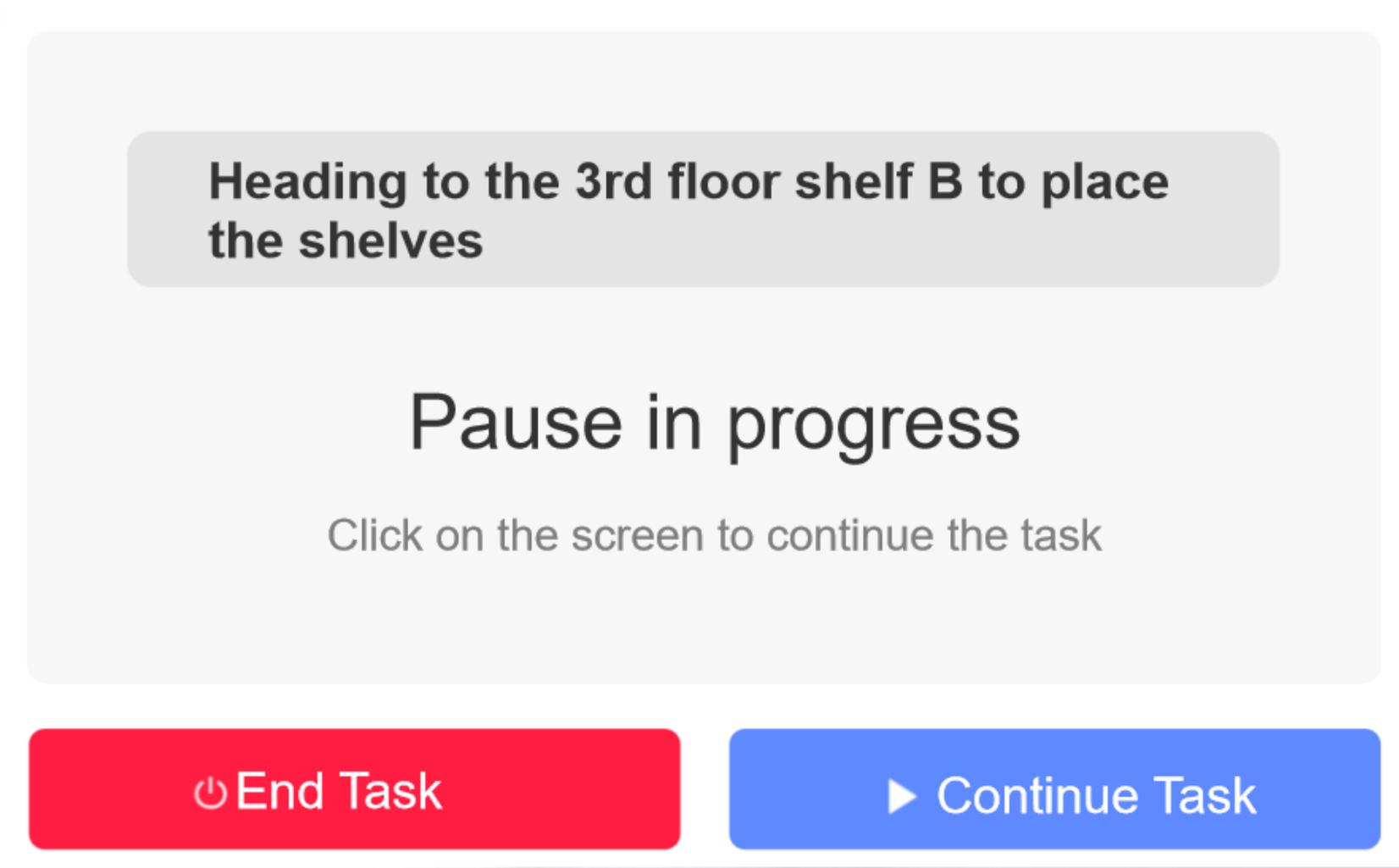
The AMR can automatically go to the jack-up point to identify and lift the shelf. After transporting the goods to the destination, it will automatically jack-down the shelf.

1. Select [Quick task] on the homepage to enter the quick task interface shown in the figure below.
2. Select the point where you want to transport the shelf in the "Jack-up point" list on the left side of the interface, and select the point where you want to place the shelf in the "Jack-down point" list on the right side.
3. After selecting the transport route, you can choose the robot action after the task is completed: "Return to the charging pile, return to the standby point, stay in place".
5. Click [Execute Now] and the robot will perform the task after the broadcast prompt.



Pause the task

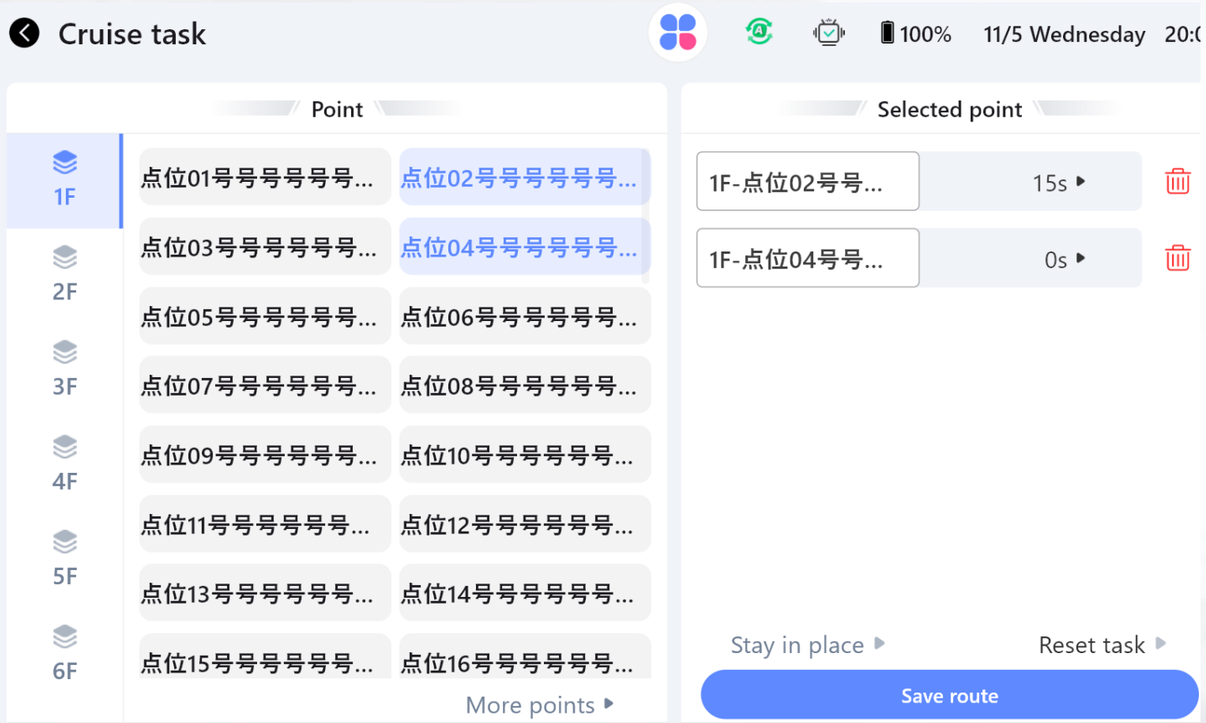
When the robot is in the process of tasks, the user can click the screen to pause the robot task. At this time, the robot is in the paused task state. Click the screen or the [Continue task] button to perform the task continuously. Click [End task] to cancel the task.

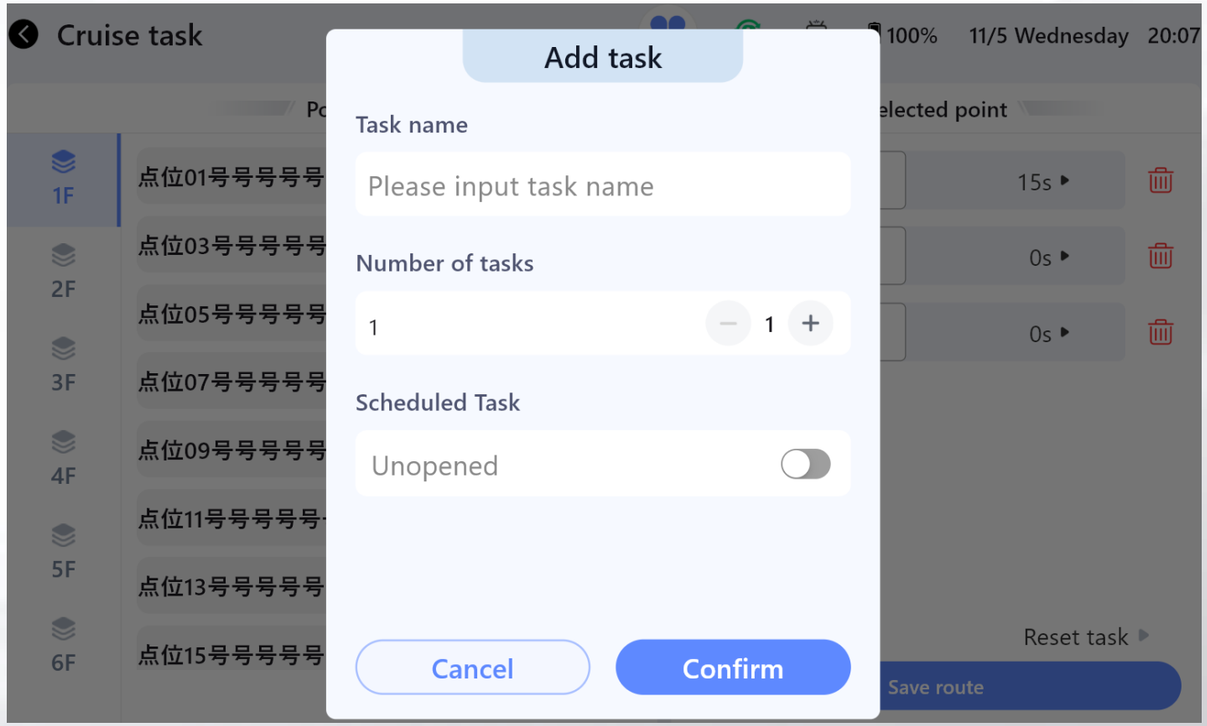


Cruise task

Deliver the goods to the designated point. After finishing all tasks, the robot returns to the designated location and waits for orders.

1. Select [Cruise task] on the homepage to enter the cruise task interface shown in the figure below.
2. Select the delivery point in the "Point" list on the left side of the interface.
3. The delivery route is displayed on the right, and the duration for each point can be configured.
4. Click [Save route] to save the currently configured route and jump to the next step of configuration.
5. Enter the task name.
6. Enter the number of times the task is to be executed.
7. You can choose whether it is scheduled task.

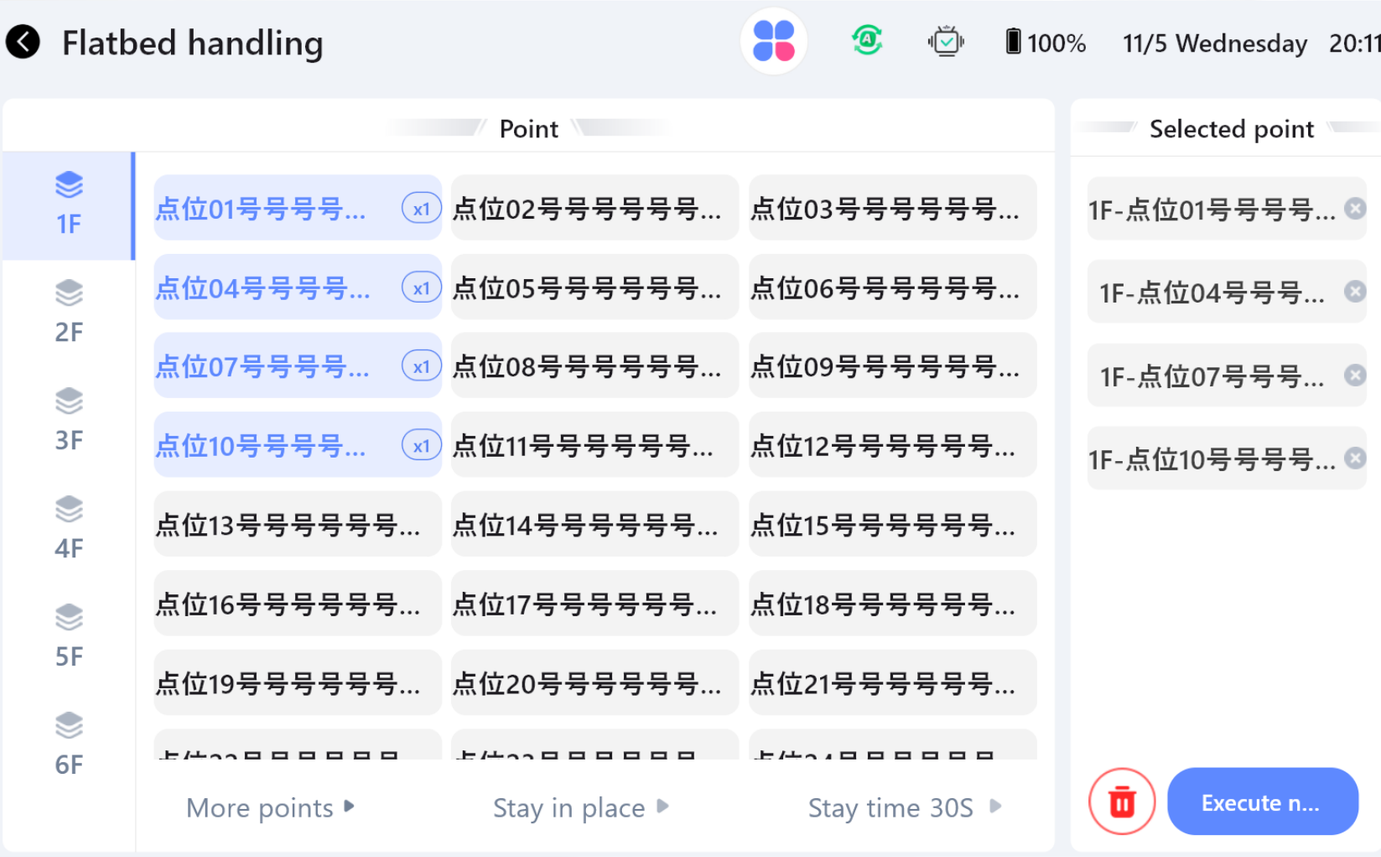




Flatbed handling

The robot can perform tasks at a single point or multiple points.

1. Select [Quick task] on the homepage to enter.
2. Select points on the interface to form a route.
3. After selecting the route, you can choose the robot action after task is completed: "Return to the charging pile, return to the standby point, stay in place"
4. You can choose the speed of transportation, low-speed, medium-speed, and high-speed.
5. Click [Execute now] and the robot will perform the task.

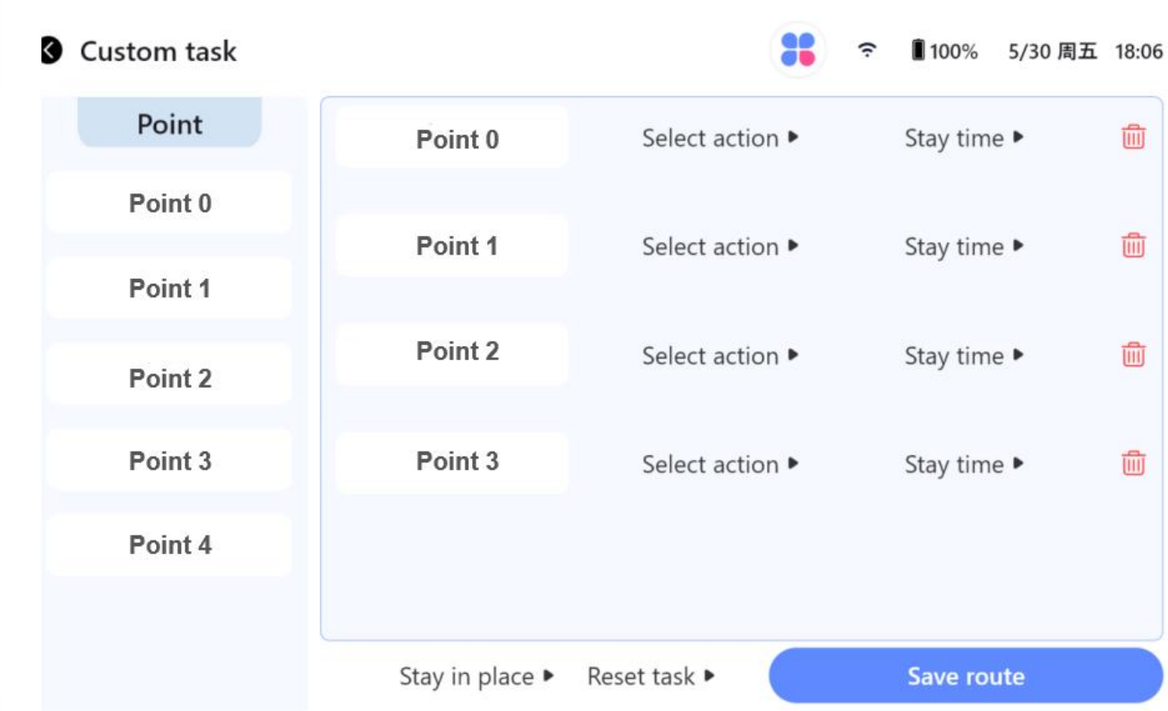


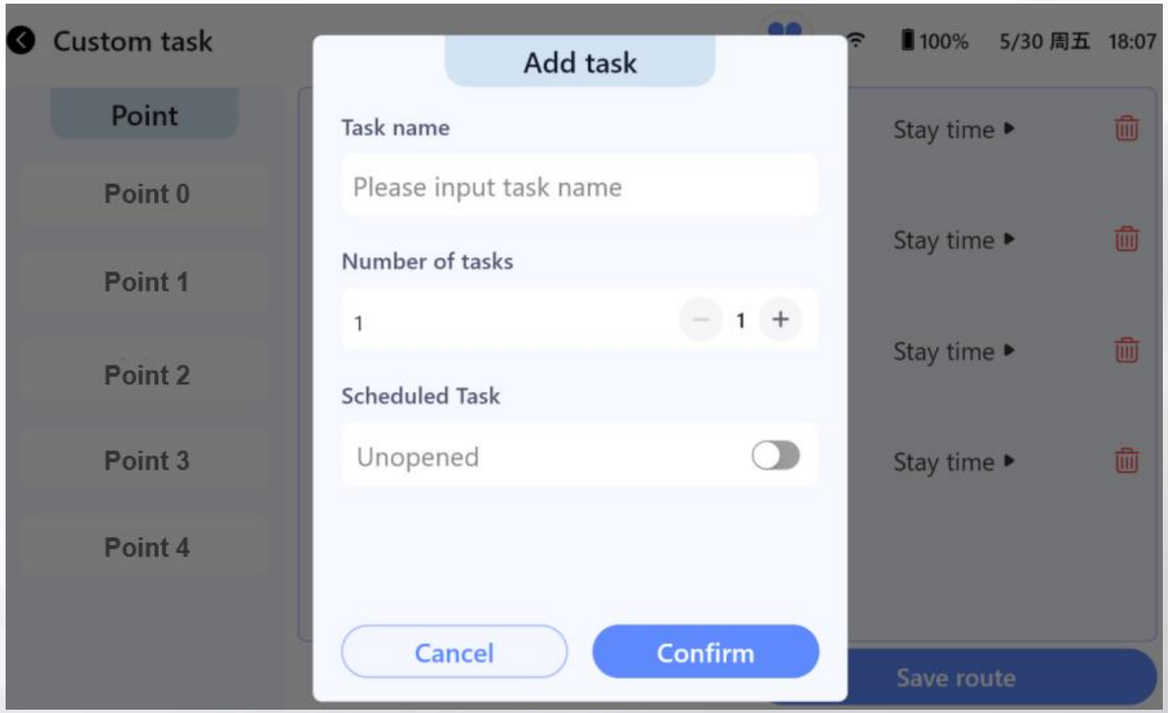
Custom task

Deliver the goods to the designated point. After finishing all tasks, the robot returns to the designated location and waits for orders.

Task configuration process

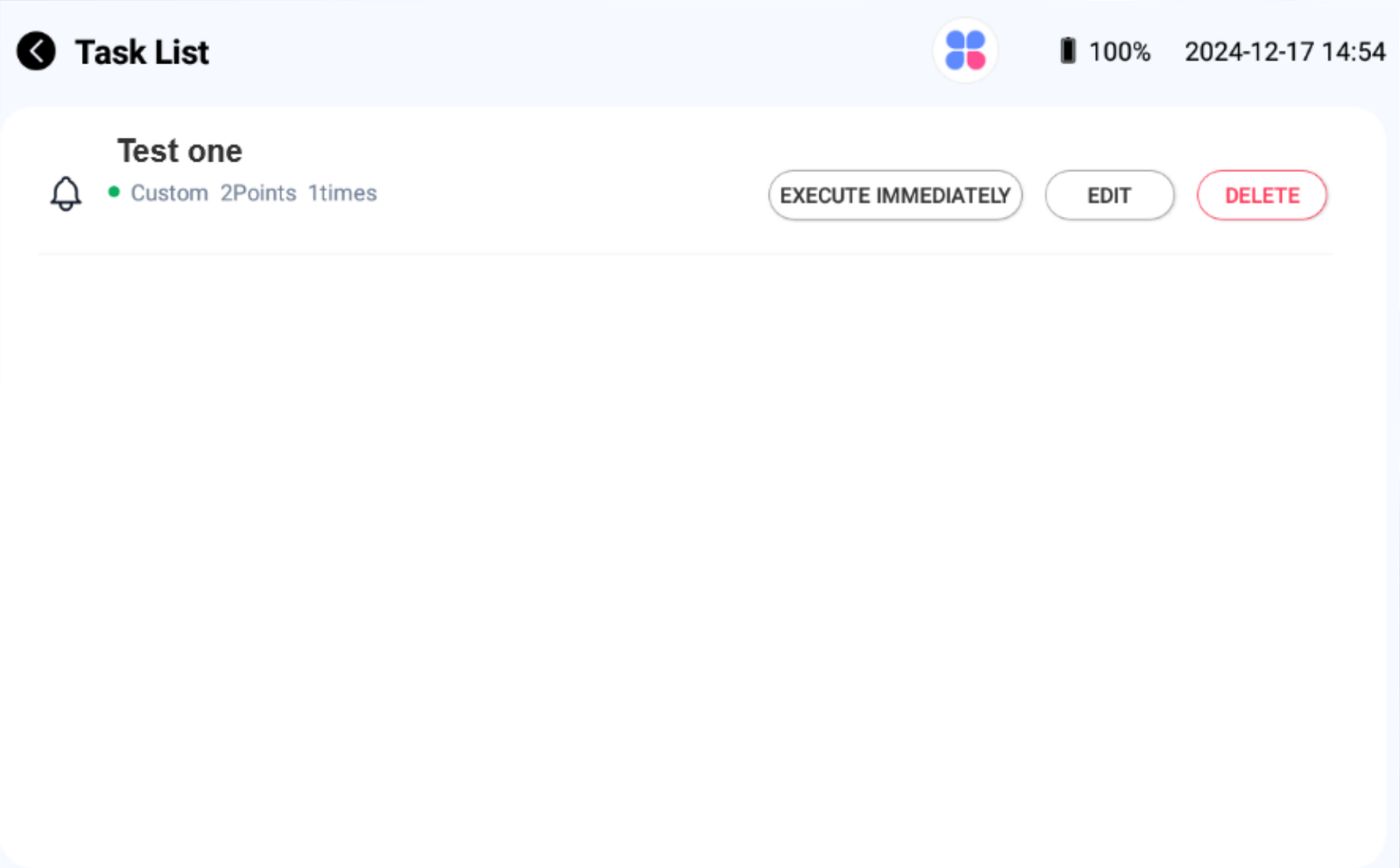
1. Select [Custom task] on the homepage to enter the custom task interface.
2. Select the delivery point in the "Point" list on the left side of the interface.
3. The delivery route is displayed on the right, and the point action (Jack-up, Jack-down, No action) and stay time of each point can be configured.
4. Click [Start now] and the robot will go to the target point along the route to perform the task.
5. Enter the task name.
6. Enter the number of times the task is to be executed.
7. You can choose whether it is scheduled task.





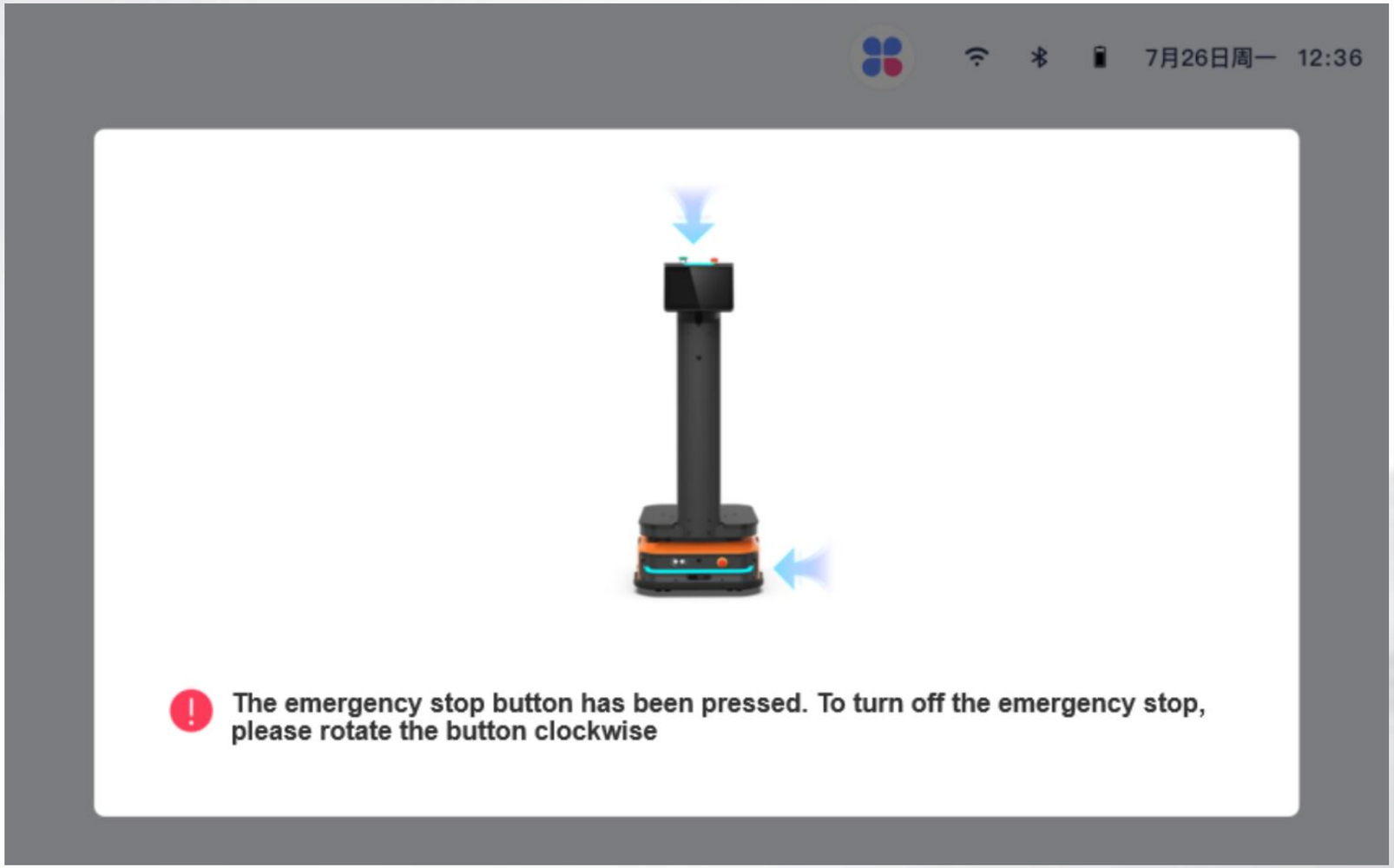
Task list

The cruise tasks and custom tasks will be displayed in the task list. This interface can be used to manage the task list.



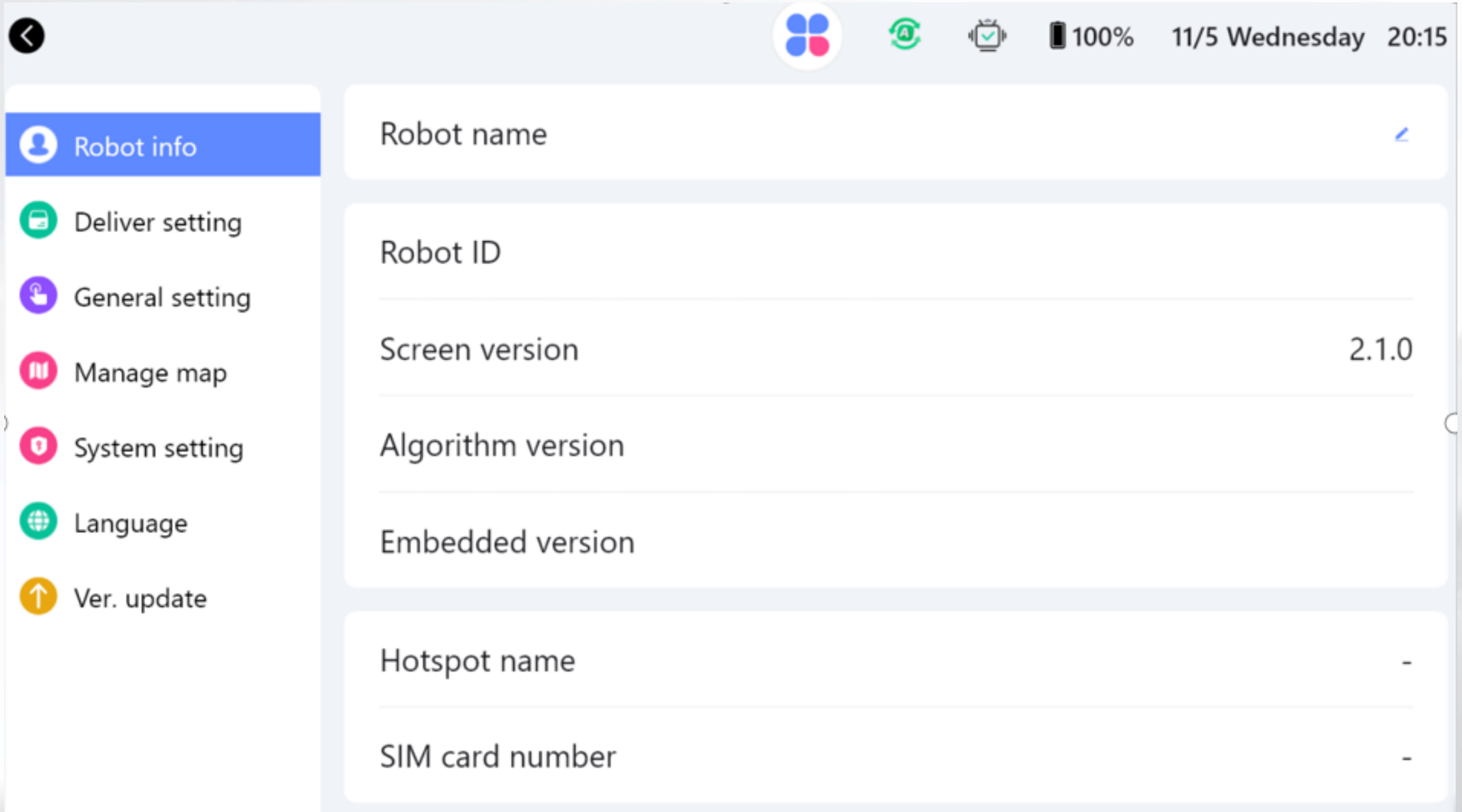
Emergency stop button

There are emergency stop buttons on the top of the robot and on the left and right sides of the chassis. In case of emergency during movement, press the emergency stop button to stop the robot. Turn the emergency stop button clockwise and follow the prompts on the interface to resume the robot's operation.

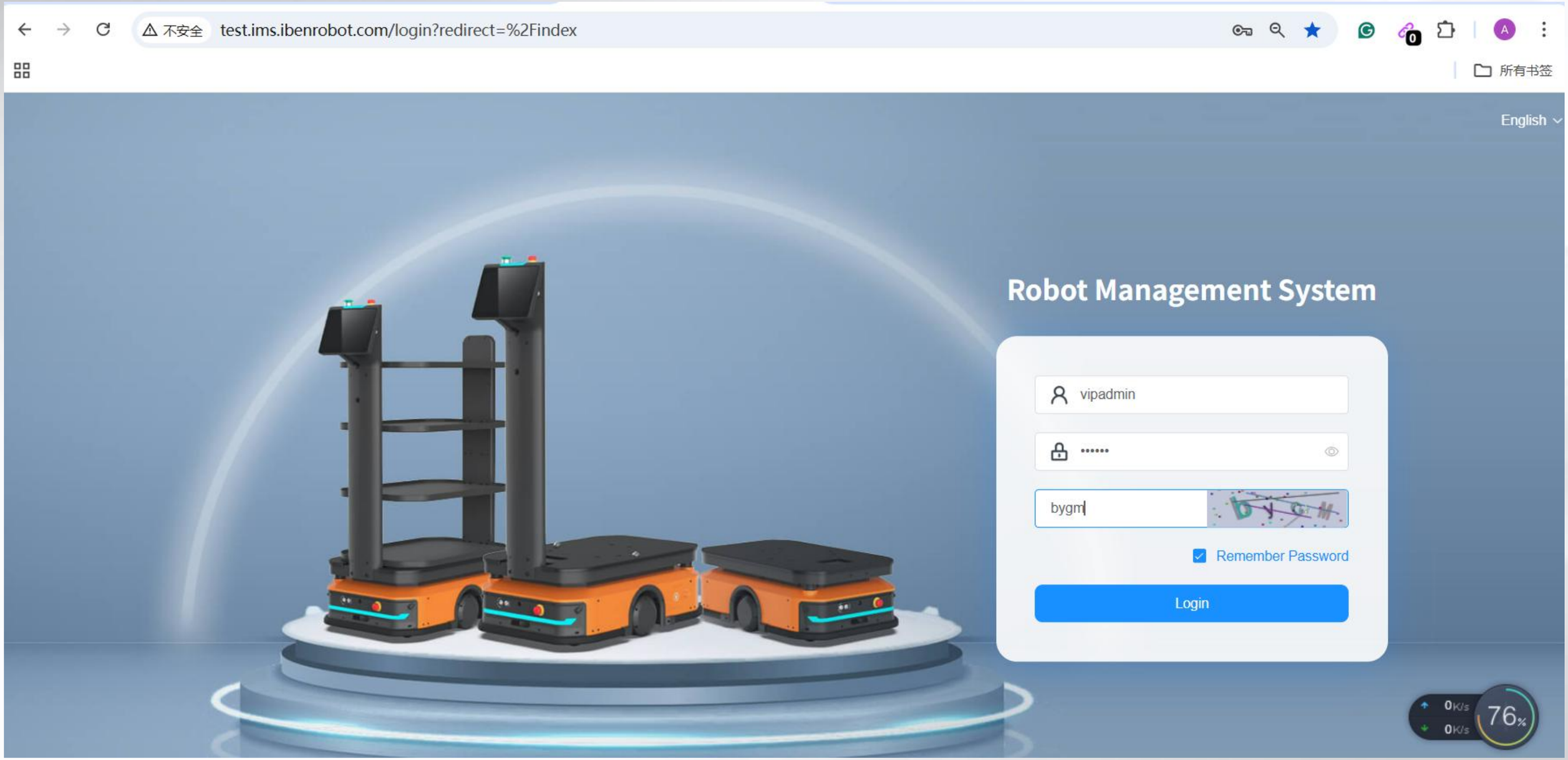


APP Setting

There is a [Setting] button on the homepage. Click it to enter the settings interface. You need to enter a password to enter the settings. The default password is **6666**.



iMS Cloub Platform Setting



Hardware

Top-viewTop-view VSLAM camera
Interactive lights
10.1 inch interactive screen
Depth camera
Front-view RGB camera
Positioning pin
Front-view RGB camera
LiDAR
Switch&brake release button
Depth camera
Emergency stop button
Fully surrounded anti-collision strip
Speaker
Power&Manual charging port
Emergency stop button
One-click release
Expanded interface
Detachable column module
Lifting module (X300B-L)
Expanded interface
LiDAR
Emergency stop button
Lithium iron phosphate battery
Turning interactive lights
Charging module
Debugging interface

Hardware

Top-view VSLAM camera
Interactive lights
10.1 inch interactive screen
Depth camera
Front-view RGB camera
Brake release button
Tray fixed position
Adjustable tray
Front-view RGB camera
LiDAR
Switch&brake release button
Depth camera
Emergency stop button
Fully surrounded anti-collision strip
Speaker
Power&Manual charging port
Emergency stop button
One-click release
Expanded interface
Detachable column module
LiDAR
Emergency stop button
Lithium iron phosphate battery
Turning interactive lights
Charging module
Debugging interface

X300 Series Parameters

Product Model	X300B-L	X300B	X300T
Picture			
Robot Size	870mm*504mm*1200mm Chassis height: 270mm Lifting stroke: 60mm	870mm*504mm*1200mm	870mm*504mm*1200mm
Load Size/Height	Standard shelf: 700mm700mm; Max shelf size: 900mm800mm; Max shelf height: 1500mm (from ground))	Pallet bearing surface: ≤700mm*500mm; Max item height: 1500mm (from chassis to item top)	

Maximum Load	**300KG**	**Charging Time**	**3.5h**
Maximum Speed	**1.0m/s (no load), 0.7m/s (loaded)**	**Obstacle Detection**	Diagonal LiDAR + 2 sets of depth cameras
Battery Life	**7h-12h**	**Operating Temperature**	0-45°C

iBEN ROBOT

THANK YOU

Contact us
Tel: 86 188 1433 8423
Email: alice@ibenrobot.cn
Official website: https://www.iBENrobot.com
Beijing headquater: No. 116, Huangchang Road, Chaoyang District, Beijing
Shenzhen operation center: 21F, Changjiang Center, Renmin Road, Longhua District, Shenzhen
Anhui manufacturer center: No. 1, Lianxing Road, Binjiang Street, Yingjiang District, Anqing City, Anhui Province
Beijing Huiwen Technology (Group) Co., LTD